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Maths puzzles for early finishers

# Early Finisher Puzzles

## Primary 6

## Semester 2

Terms 3 and 4

20 single-page puzzles
Almost every puzzle checks itself
One star and two star pages
Teacher answer key included

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# Teacher notes (DRAFT)

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## What this is

Twenty single-page maths puzzles for Primary 6 pupils who finish their work early. This is Semester 2 of two booklets (Terms 3 and 4). Each page stands alone, so you can hand out one page or keep a tray of pages. The pupil needs no help to start.

## Which terms

The pages are grouped by the order in which Primary 6 topics are typically taught: this booklet follows Terms 3 and 4, the other booklet follows Semester 1 (Terms 1 and 2). Schools vary, so check your scheme of work: use a page whenever its topic has been taught, from either booklet.

## Stars

One star pages practise a topic as it is usually taught (10 pages). Two star pages use the same topics with two steps or two clues to combine (10 pages).

## How the pages check themselves

The boxed strip at the foot of each page tells the pupil how to check. The check shows that something is wrong. It does not give the answer, and it asks the pupil to see whether the answer makes sense. Every answer box and every letter box must be filled: a box left blank gives no letter, and the check cannot see it.

- **Code word.** The answers give letters that spell a word. The strip gives no hint of the word. Each word can be spelt from its letters in one way only, so answers in the wrong boxes never make another word, and one or two wrong answers very rarely do. A box left empty gives no letter and is not seen by the check.
- **Colour by code.** Each answer is found in a key of five numbers and the part is coloured with that crayon. The pupil counts the parts of each colour and compares with the box on the page; matching parts must share a colour.
- **Route.** The line must reach FINISH through the stated number of boxes, and every box on it must follow the rule on the page.
- **Letters in an order, a grid or a choice.** Each number, card, picture or choice carries a letter; the letters, read in the order asked, must spell a word. Where one item is left out, its letter is a decoy that is not part of the word.
- **Loop.** The numbers on touching tiles must match all the way round the loop.

Marking is not needed. To spot-check, look only at the check strip.

## How the checks were tested

A script tried every single wrong answer and every wrong arrangement on every page. Each check rejected at least 95% of them. A wrong answer that spells any of about 207 000 English words of two to nine letters counts as not rejected. Two wrong answers at once were not all tested.

# Teacher notes: using the pages (DRAFT)

## Using the pages

- **Time.** Most pages take 5 to 15 minutes; the colour-by-code pages (S2-12, S2-20) take longer because pupils colour a picture. Pupils need their own crayons (five colours: yellow, red, blue, green, orange). For a pencil-only copy the pupil writes Y, R, B, G or O in each part; the tally and the matching parts are checked the same way. Light crayons work best: a dark crayon hides the number in the part on a photocopy.
- **If the check fails.** The pupil goes back, looks for the slip, and asks a friend or you if still stuck.
- **Problem solving.** Some pages (S2-19) practise the kinds of question that appear in school tests, such as average with a changed value, tank and tap stories and questions that mix several topics, rather than one numbered syllabus outcome. The key tags them problem solving.
- **Stars.** One star: one skill, as usually taught. Two stars: the same topics with two steps or two clues to combine.
- **Drawings.** Pictures of circles, solids, graphs and angle figures are original drawings, drawn accurately to help the pupil see the shape. Pupils should work out the answer, not measure it. Pictures are not actual size.
- **Notation.** Ratios are written 3 : 5 and answers in simplest form. Where a page uses π it says whether to take 3.14 or 22/7 (22 over 7). Letters such as n, p and a stand for numbers; there are no brackets with letters. Money is written $12.50. Litres are written ℓ and cube centimetres cm³.
- **Rough paper.** Many pages need working that does not fit in the boxes. Give the pupil a sheet of rough paper to keep with the page.
- **Calculators.** Every number is chosen so that the working can be done by hand; a pupil may use a calculator if your school allows it, but the check does not need one.
- **Revision pages.** Pages whose key tag begins with Revisits practise outcomes from Primary 4 or 5 that Primary 6 pupils meet again in revision. Primary 6 content only is practised on every other page.

## Photocopying

All pages are black and white with solid lines, so they copy cleanly. Use normal toner, not a light setting. Each page is one side only. Pupils write on the page: the boxes are large. Use A4 and print at 100% (actual size).

Pictures are not actual size.

# Contents: Semester 2


| Page | Code | Puzzle | Tier |
|---|---|---|---|
| S2-01 | P6-21 | Shaded Shapes | 1 star |
| S2-02 | P6-22 | Perimeter Pick | 2 stars |
| S2-03 | P6-23 | Cuboid Corners | 1 star |
| S2-04 | P6-24 | Roots and Edges | 1 star |
| S2-05 | P6-25 | Base and Height | 2 stars |
| S2-06 | P6-26 | Tank Stories | 2 stars |
| S2-07 | P6-27 | Angles in Figures | 1 star |
| S2-08 | P6-28 | More Angles | 2 stars |
| S2-09 | P6-29 | Angle Match | 1 star |
| S2-10 | P6-30 | Average | 1 star |
| S2-11 | P6-31 | Missing Value | 2 stars |
| S2-12 | P6-32 | Colour the Butterfly | 2 stars |
| S2-13 | P6-33 | Pie Chart | 2 stars |
| S2-14 | P6-34 | Net or Not | 1 star |
| S2-15 | P6-35 | Taps and Rates | 2 stars |
| S2-16 | P6-36 | Mixed Numbers | 1 star |
| S2-17 | P6-37 | Decimal Maze | 1 star |
| S2-18 | P6-38 | GST and Discount | 1 star |
| S2-19 | P6-39 | Mixed Bag | 2 stars |
| S2-20 | P6-40 | Colour the Elephant | 2 stars |


Pages are independent. The footer code shows the page in this booklet and, in brackets, its number in the full set of 40 (P6-01 to P6-40).

## What the booklet covers

**Fractions 1.2** dividing a whole number or a proper fraction by a proper fraction **S2-19, S2-20**

**Ratio 1.3** dividing a quantity in a given ratio **S2-19, S2-20**

**Algebra 1.4** evaluating a simple expression by substitution **S2-19, S2-20**

**Algebra 1.5** solving simple linear equations with whole-number coefficients **S2-20**

**Area and Volume 1.2** area and perimeter of a semicircle and a quarter circle **S2-02**

**Area and Volume 1.3** composite figures made of squares, rectangles, triangles, semicircles and quarter circles **S2-01, S2-02**

**Area and Volume 2.1** finding a missing dimension of a cuboid given its volume and the other dimensions **S2-03, S2-12**

**Area and Volume 2.2** finding the edge of a cube given its volume **S2-04, S2-05, S2-12**

**Area and Volume 2.3** finding the height of a cuboid given its volume and base area **S2-05, S2-06, S2-12**

**Area and Volume 2.4** finding the area of a face of a cuboid given its volume and one dimension **S2-05**

**Area and Volume 2.5** use of the square-root and cube-root signs **S2-04, S2-19**

**Geometry 1.1** finding unknown angles in composite figures (square, rectangle, triangle, parallelogram, rhombus, trapezium), without additional construction **S2-07, S2-08, S2-09**

**Statistics 1.1** average as total value ÷ number of data **S2-10, S2-12, S2-19, S2-20**

**Statistics 1.2** relationship between average, total value and number of data **S2-11, S2-12**

**Revisits Primary 4 Statistics** reading and interpreting pie charts **S2-13**

**Revisits Primary 4 Geometry** identifying nets of a cube and a cuboid **S2-14**

**Revisits Primary 4 Decimals** adding, subtracting, multiplying and dividing decimals **S2-17**

**Revisits Primary 5 Fractions** adding and subtracting mixed numbers; multiplying a fraction by a whole number or a fraction **S2-16**

**Revisits Primary 5 Percentage** a percentage part of a whole; discount, GST and annual interest **S2-13, S2-18, S2-19, S2-20**

**Revisits Primary 5 Rate** rate as the amount of one quantity per unit of another; finding the rate, the total or the number of units **S2-15**

**Revisits Primary 5 Volume** volume of a cuboid; volume of liquid in a rectangular tank; litres and $cm^3$ **S2-06, S2-15, S2-19**

Outcomes practised in the other booklet: Fractions 1.1; Percentage 1.1; Percentage 1.2; Ratio 1.1; Ratio 1.2; Ratio 1.4; Ratio 1.5; Ratio 1.6; Ratio 1.7; Algebra 1.1; Algebra 1.2; Algebra 1.3; Area and Volume 1.1.